

UNDECIDABILITY

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AP/CSE

$RL \rightarrow ReL$

TM

LEARNING OBJECTIVE

- To Design Turing machines for any Languages (K3)
 - To Understand the concept of Turing Machine

REVISIT - TM

- A Turing machine (TM) is a 7-tuple

$M = (Q \cup \{h_a, h_r\}, \Sigma, \Gamma, \delta, q_0, B, F)$ where

- Q – A finite set of states of the finite control. $Q + \underline{h_a}$ and $\underline{h_r}$
- Σ – A finite set of input symbols
- Γ – A set of tape symbols, with Σ being a subset
- q_0 – The start state, in Q
- B – The blank symbol in Γ , *not* in Σ (should not be an input symbol)
- F – The set of final or accepting states

$$(\underline{Q}, \underline{\Sigma}, \underline{\Gamma}, \underline{\delta}, \underline{B}, \underline{q_0}, \underline{F})$$

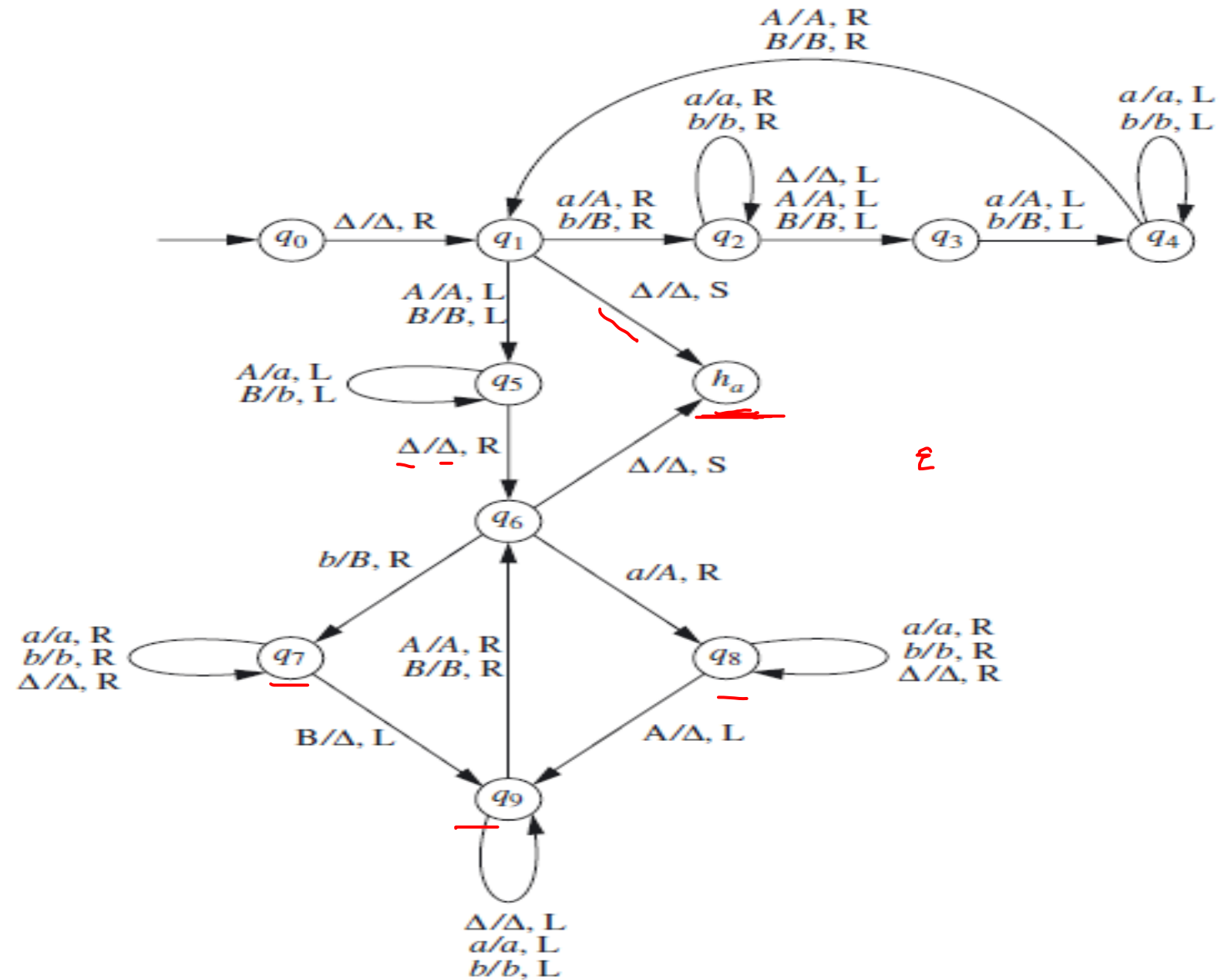
$$L = \{0^n 1^n / n \geq 1\}$$

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EXAMPLE

$$L = \{\underline{xx} \mid x \in (a,b)^*\}$$



PARSING EXAMPLE WITH H_A, H_R

$q_0 \Delta \underline{aba}$	$\vdash \Delta q_1 aba$ $\vdash \Delta A q_4 b A$ $\vdash \Delta A \underline{q_3} B A$	$\vdash \Delta A q_2 ba$ $\vdash \Delta q_4 A b A$ $\vdash \Delta A h_r B A$	$\vdash^* \Delta A b a q_2 \Delta$ $\vdash \Delta A q_1 b A$ (reject)	$\vdash \Delta A b q_3 a$ $\vdash \Delta A B q_2 A$
$q_0 \Delta \underline{ab}$	$\vdash \Delta q_1 ab$ $\vdash \Delta q_4 A B$ $\vdash \Delta q_6 a B$	$\vdash \Delta A q_2 b$ $\vdash \Delta A q_1 B$ $\vdash \Delta A \underline{q_8} B$	$\vdash \Delta A b q_2 \Delta$ $\vdash \Delta q_5 A B$ $\vdash \Delta A h_r B$	$\vdash \Delta A q_3 b \Delta$ $\vdash q_5 \Delta a B$ (reject)
$q_0 \Delta \underline{aa}$	$\vdash \Delta q_1 aa$ $\vdash \Delta q_4 A A$ $\vdash \Delta q_6 a A$ $\vdash \Delta A \underline{h_a} \Delta$	$\vdash \Delta A q_2 a$ $\vdash \Delta A q_1 A$ $\vdash \Delta A q_8 A$ (accept)	$\vdash \Delta A a q_2 \Delta$ $\vdash \Delta q_5 A A$ $\vdash \Delta q_9 A$	$\vdash \Delta A q_3 a \Delta$ $\vdash q_5 \Delta a A$ $\vdash \Delta A q_6 \Delta$

RECURSIVE (R) AND RECURSIVELY ENUMERABLE(RE) LANGUAGES

TM $\begin{cases} RL \\ \underline{ReL} \end{cases}$

DECIDABILITY VS. UNDECIDABILITY

- There are two types of TMs (based on halting):

(Recursive)

$L = \{0^n 1^n 2^n \mid n \geq 1\} \xrightarrow{RT} TM \rightarrow \text{halts}$

TMs that *always* halt, no matter accepting or non-accepting $\equiv$
DECIDABLE PROBLEMS

(Recursively enumerable)

$A \rightarrow A$ M or may not halt $\rightarrow \underline{R}$

TMs that *are guaranteed to halt only on acceptance*. If non-accepting, it may or may not halt (i.e., could loop forever).

- Undecidability:

reject

– Undecidable problems are those that are not recursive

$R \neq L$ cannot const TM

RECURSIVE LANGUAGE

- A Language L is Recursive if and only if there is a TM that decides L.
 - Let $M = (Q, \Sigma, \Gamma, \delta, q_0, B, \underline{H})$ such that
 - $H = \{h_a, h_r\}$
 - $L \subseteq \Sigma^*$ Is a language
 - Assume that the initial configuration of the TM is (q_0, w)
 - M decides L if, for all strings $w \in \Sigma^*$
 - Either $w \in L$, in which case M accepts w
 - Or $w \notin L$, then M rejects w

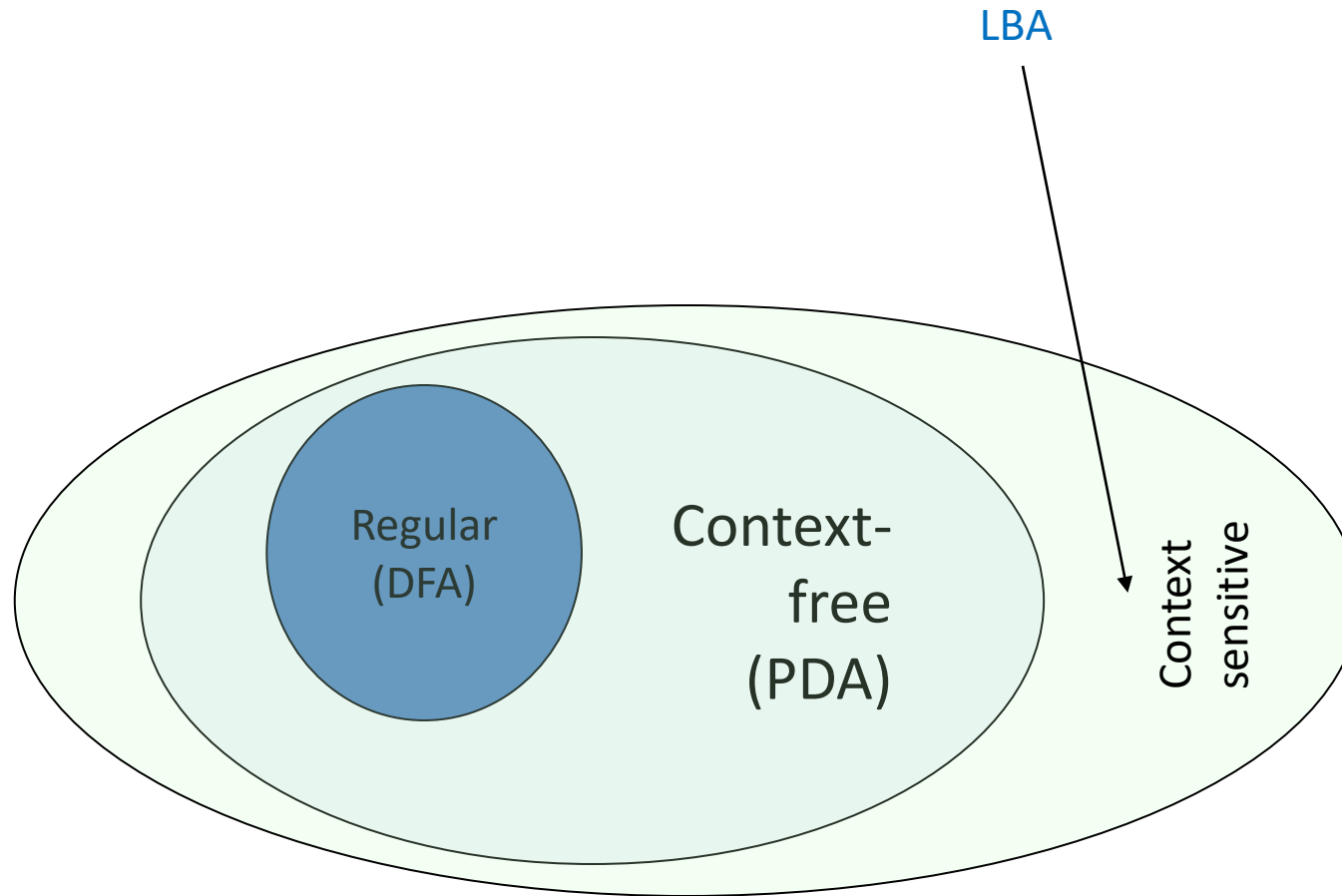
$L \rightarrow \underline{TM}$ | $w \in L \quad TM \rightarrow \text{Accept}$
 $w \notin L \quad TM \rightarrow \text{Reject}$

RECURSIVE ENUMERABLE LANGUAGE

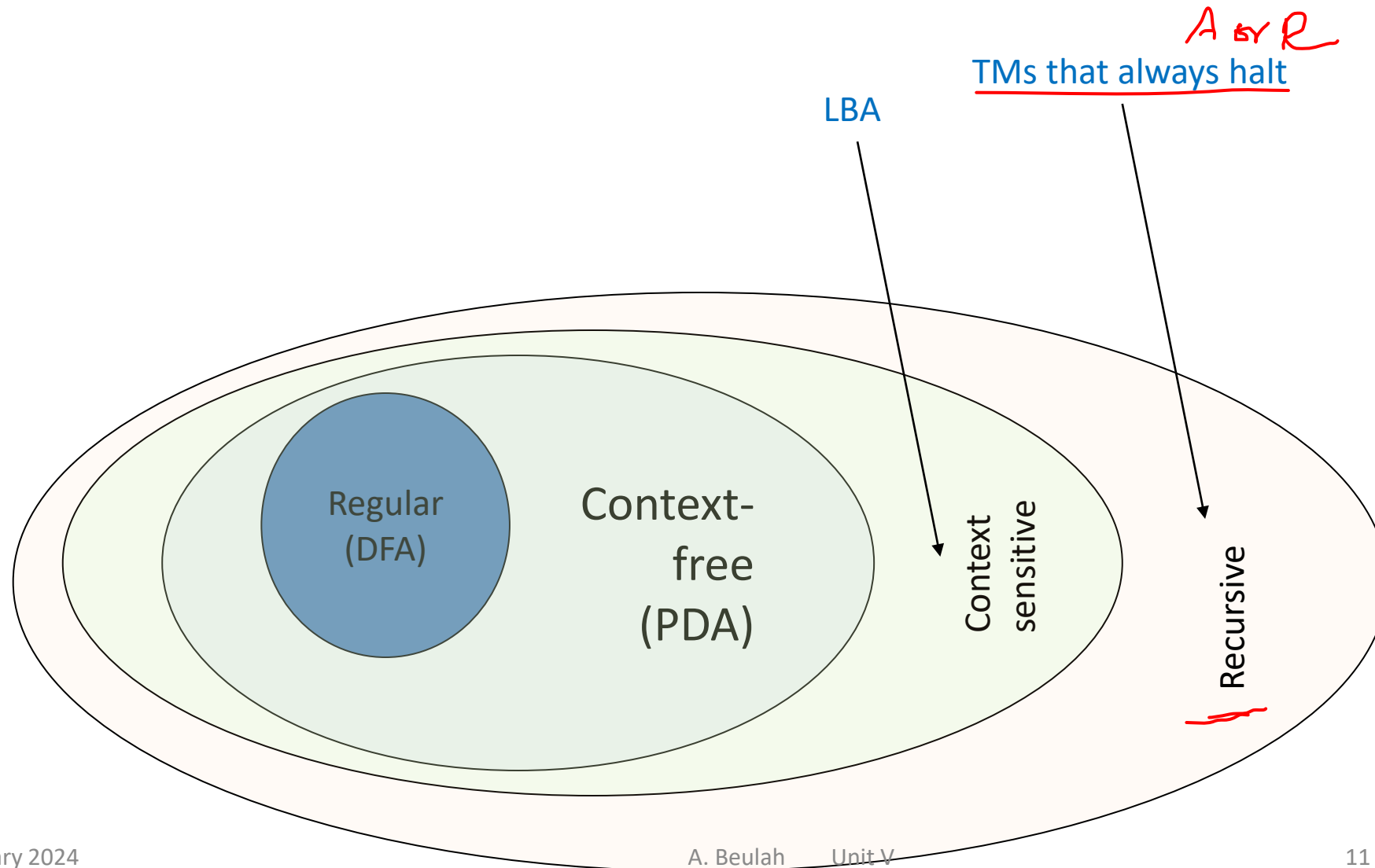
- A Language L is Recursive Enumerable if and only if there is a TM that semidecides L .
 - Let $M = (Q, \Sigma, \Gamma, \delta, q_0, B, H)$ such that
 - $H = \{h_a, h_r\}$
 - $L \subseteq \Sigma^*$ is a language
 - Assume that the initial configuration of the TM is (q_0, w)
 - M semidecides L if, for all strings $w \in \Sigma^*$
 - Either $w \in L$, in which case M accepts w
 - Or $w \notin L$, then M does not halt

$L \rightarrow TM_M$
 $w \in L \quad M \rightarrow \text{Accept} \rightarrow \text{Halt}$
 $w \notin L \quad M \rightarrow \text{Reject} \rightarrow \text{Halt}$

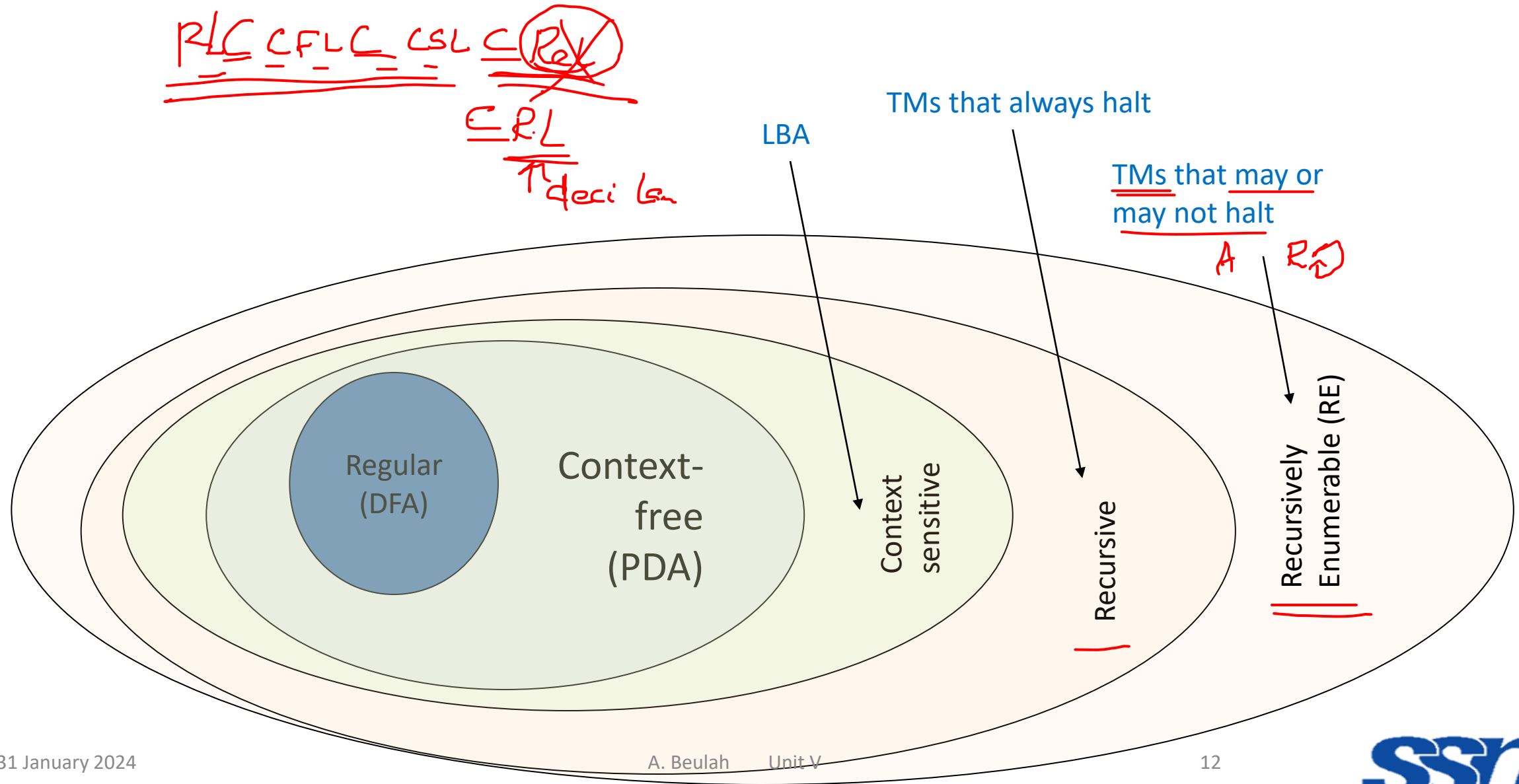
RECURSIVE, RE, UNDECIDABLE LANGUAGES



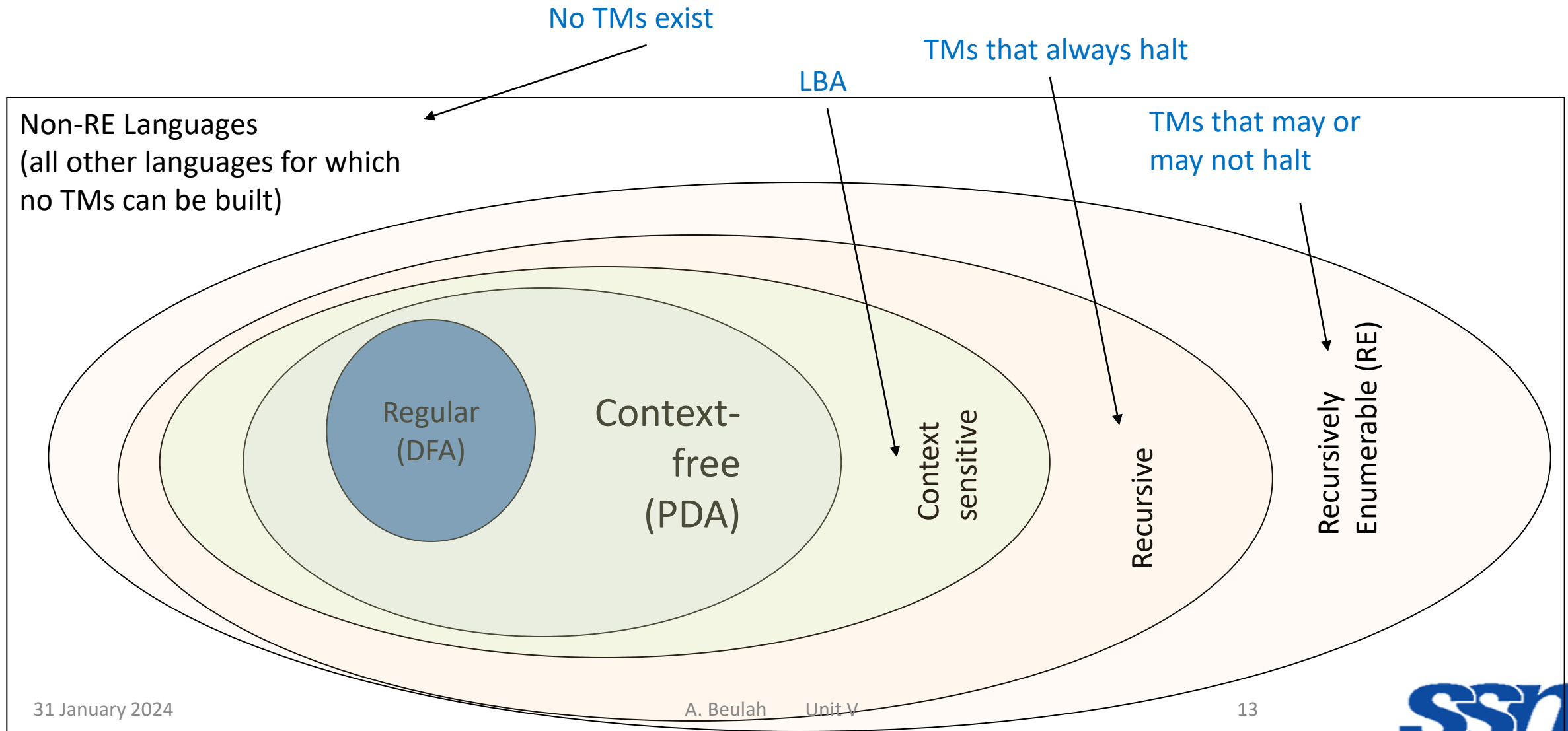
RECURSIVE, RE, UNDECIDABLE LANGUAGES



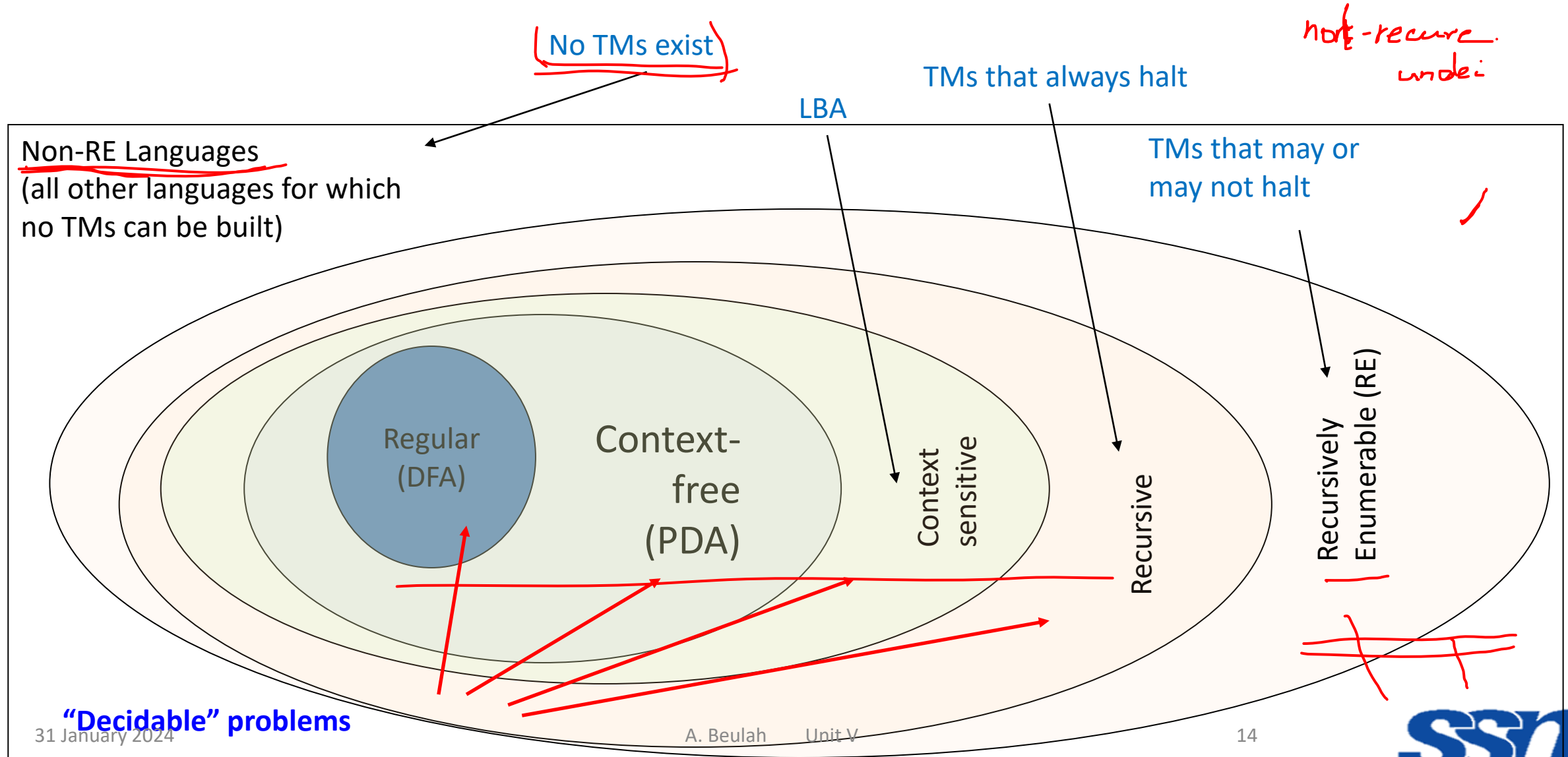
RECURSIVE, RE, UNDECIDABLE LANGUAGES



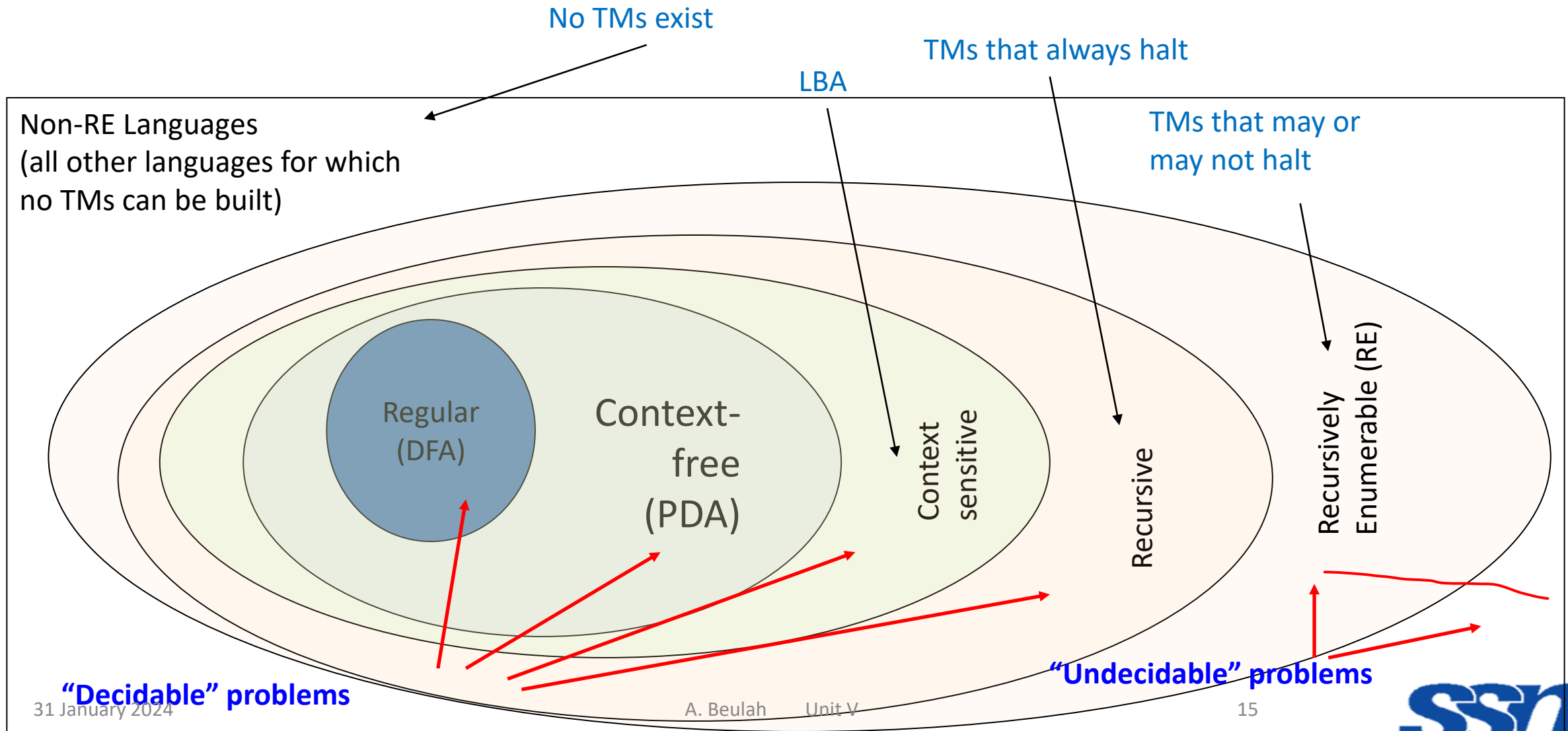
RECURSIVE, RE, UNDECIDABLE LANGUAGES



RECURSIVE, RE, UNDECIDABLE LANGUAGES

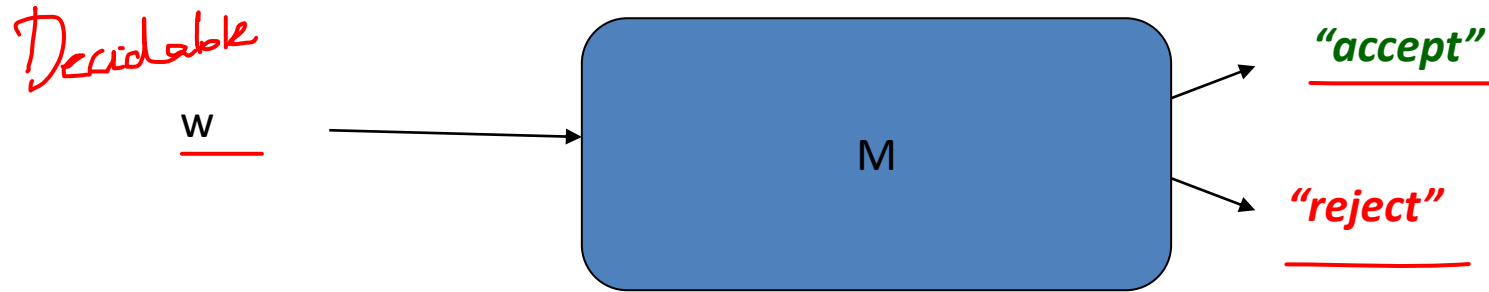


RECURSIVE, RE, UNDECIDABLE LANGUAGES

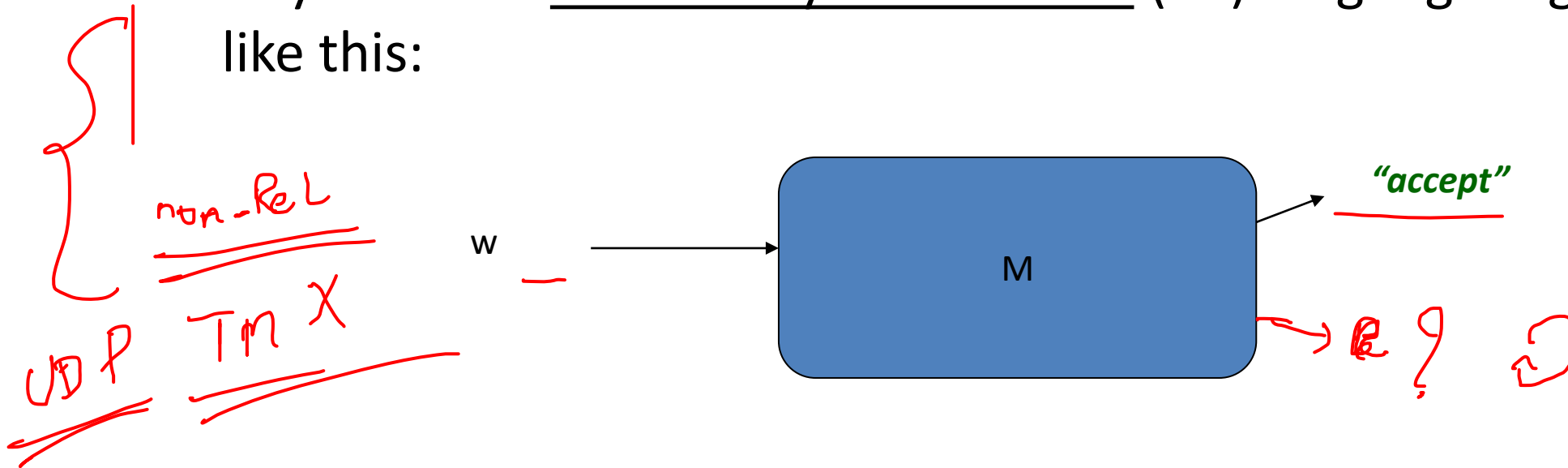


RECURSIVE, RE, UNDECIDABLE LANGUAGES

- Any TM for a Recursive language is going to look like this:



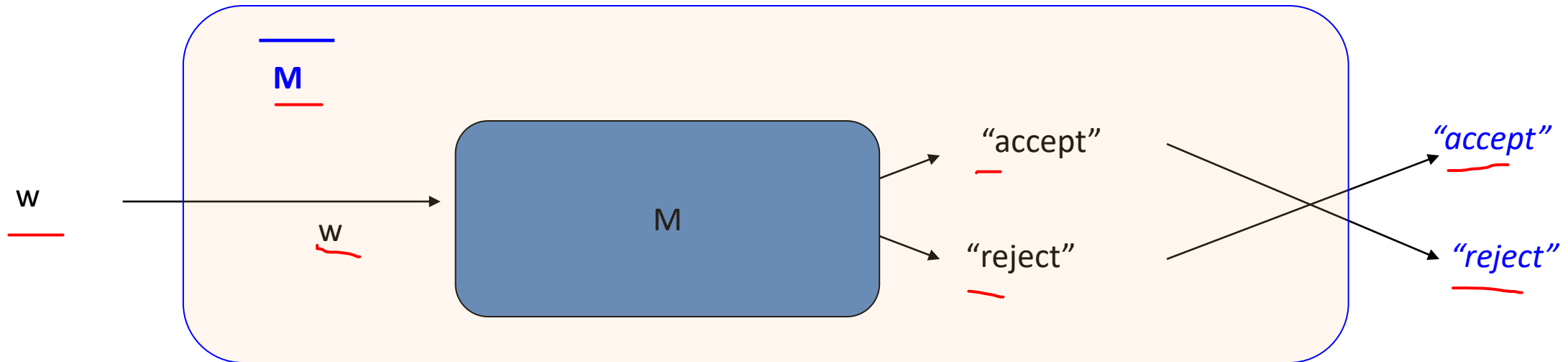
- Any TM for a Recursively Enumerable (RE) language is going to look like this:



CLOSURE PROPERTIES OF RECURSIVE (R) AND RECURSIVELY ENUMERABLE(RE) LANGUAGES

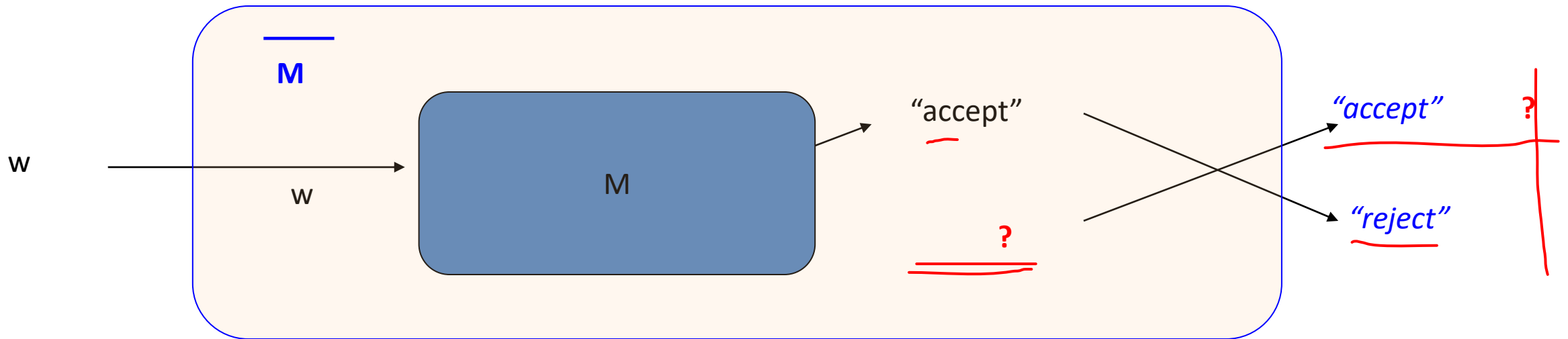
RL ARE CLOSED UNDER COMPLEMENTATION

- If L is Recursive, $\overline{L}$ is also Recursive



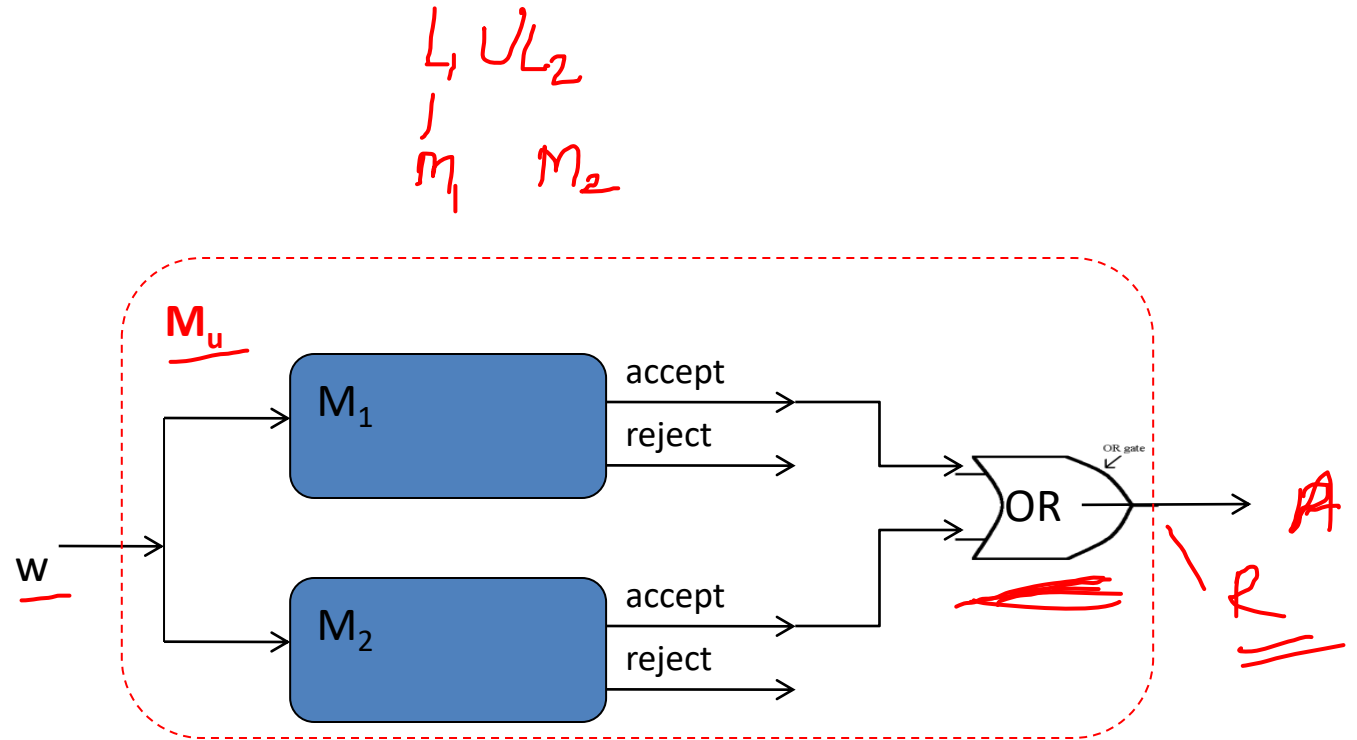
ARE ~~RL~~ ^{RE} ~~CLOSED~~ UNDER COMPLEMENTATION? (NO)

- If L is RE, $\overline{L}$ need not be RE



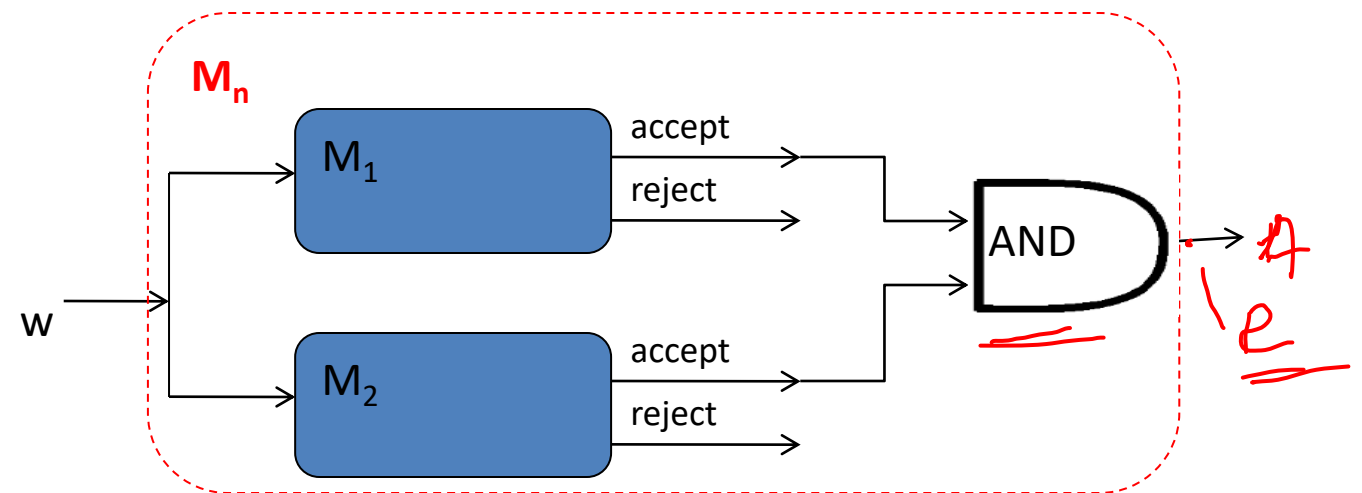
RECURSIVE LANGS ARE CLOSED UNDER UNION

- Let $M_u = \text{TM for } L_1 \cup L_2$
- M_u construction:
 1. Make 2-tapes and copy input w on both tapes
 2. Simulate M_1 on tape 1
 3. Simulate M_2 on tape 2
 4. If either M_1 or M_2 accepts, then M_u accepts
 5. Otherwise, M_u rejects.



RL ARE CLOSED UNDER INTERSECTION

- Let $M_n = \text{TM for } L_1 \cap L_2$
- M_n construction:
 1. Make 2-tapes and copy input w on both tapes
 2. Simulate M_1 on tape 1
 3. Simulate M_2 on tape 2
 4. If either M_1 AND M_2 accepts, then M_n accepts
 5. Otherwise, M_n rejects.



OTHER CLOSURE PROPERTY RESULTS

- Recursive languages are also closed under:
 - Concatenation —
 - Kleene closure (star operator) —
 - Homomorphism, and inverse homomorphism
- RE languages are closed under:
 - Union, intersection, concatenation, Kleene closure

- RE languages are *not* closed under:
 - complementation /

TEST YOUR KNOWLEDGE

- Let L_1 be a recursive language. Let L_2 and L_3 be languages that are recursively enumerable but not recursive. Which of the following statements is not necessarily true?
 - $L_2 - L_1$ is recursively enumerable.
 - $L_1 - L_3$ is recursively enumerable
 - $L_2 \cap L_1$ is recursively enumerable
 - $L_2 \cup L_1$ is recursively enumerable

SUMMARY

- What is undecidability
- Recursive and Recursive enumerable languages

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008